Defending Agentic Federated Learning Systems Against Explanation Poisoning

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Problem Statement and Objectives

Problem Statement:

Agentic Federated Learning is vulnerable to **Explanation Poisoning**, where adversaries hide backdoors behind fake, plausible explanations (e.g., manipulated heatmaps). Current defenses fail because they audit **weights** (Euclidean distance) rather than **logic**. This allows "Stealth Attacks" to bypass statistical checks, creating models that are accurate but causally corrupt.

Objectives:

- **Formalize Logic-Based Governance:** Shift auditing from Parameter Space (Weights) to Topological Space (Causal Logic).
- **Defeat Explanation Poisoning:** Prove that minimizing Graph Edit Distance creates a mathematical paradox that forces adversaries to disarm their backdoors.
- **Enable Scalable Verification:** Implement Neural Graph Matching to approximate NP-Hard graph validations in real-time.

Literature Review

Paper / Work	Authors	Venue	Methodology	Advantages	Disadvantages / Limitations
Scaffolding Attacks on Explainable ML	Liyanage et al.	IEEE (2024)	Introduced *Explanation Scaffolding*, where model behavior is biased while post-hoc explainers (e.g., SHAP) are manipulated to conceal the bias	Demonstrates that explanation-level manipulation is possible without degrading accuracy	Does not propose any concrete or mathematical defense; relies on post-hoc auditing assumptions
Output Shuffling Attacks on SHAP	Jiawang Liu, Changgen Peng, Weijie Tan, Chenghui Shi	NeurIPS (2024)	Wrapped the target model with a surrogate function that preserves predictions but alters SHAP attributions	Shows that SHAP can be systematically fooled while protected attributes remain influential	Explains the vulnerability but offers no mitigation strategy beyond "stronger auditing"
Federated Backdoor Attacks in Clinical AI	Jun Yuan, Aritra Dasgupta	Nature Medicine (2026)	Empirical study showing that small-scale poisoned updates can implant backdoors in large federated medical models	Highlights extreme real-world risk in healthcare FL systems	Focuses on attack feasibility; explanation integrity and trust mechanisms are not addressed
FedREDefense	Yueqi Xie, Minghong Fang , Neil Zhenqiang Gong	ICML (2024)	Used autoencoder reconstruction error to detect anomalous client updates	Effective against noisy, unstructured, or high-variance attacks	Fails against clean, structured attacks such as Explanation Poisoning

Literature Gap

Despite advances in federated learning security, explainability, and causal modelling, a critical gap exists at their intersection:

1. Existing federated defences verify numerical similarity or output correctness but do not validate internal reasoning.
2. Explanation-based auditing assumes explanations are faithful, an assumption violated by explanation poisoning.
3. Causal reasoning is used for interpretability and fairness but not for adversarial detection.
4. No existing framework enforces consensus over reasoning logic across federated agents.

As a result, federated systems lack a mechanism to distinguish benign autonomy from malicious semantic deviation.

Proposed Solution - Proof of Reasoning

Conceptual Shift: Moving from "What the model predicts" to "Why the model predicts it."

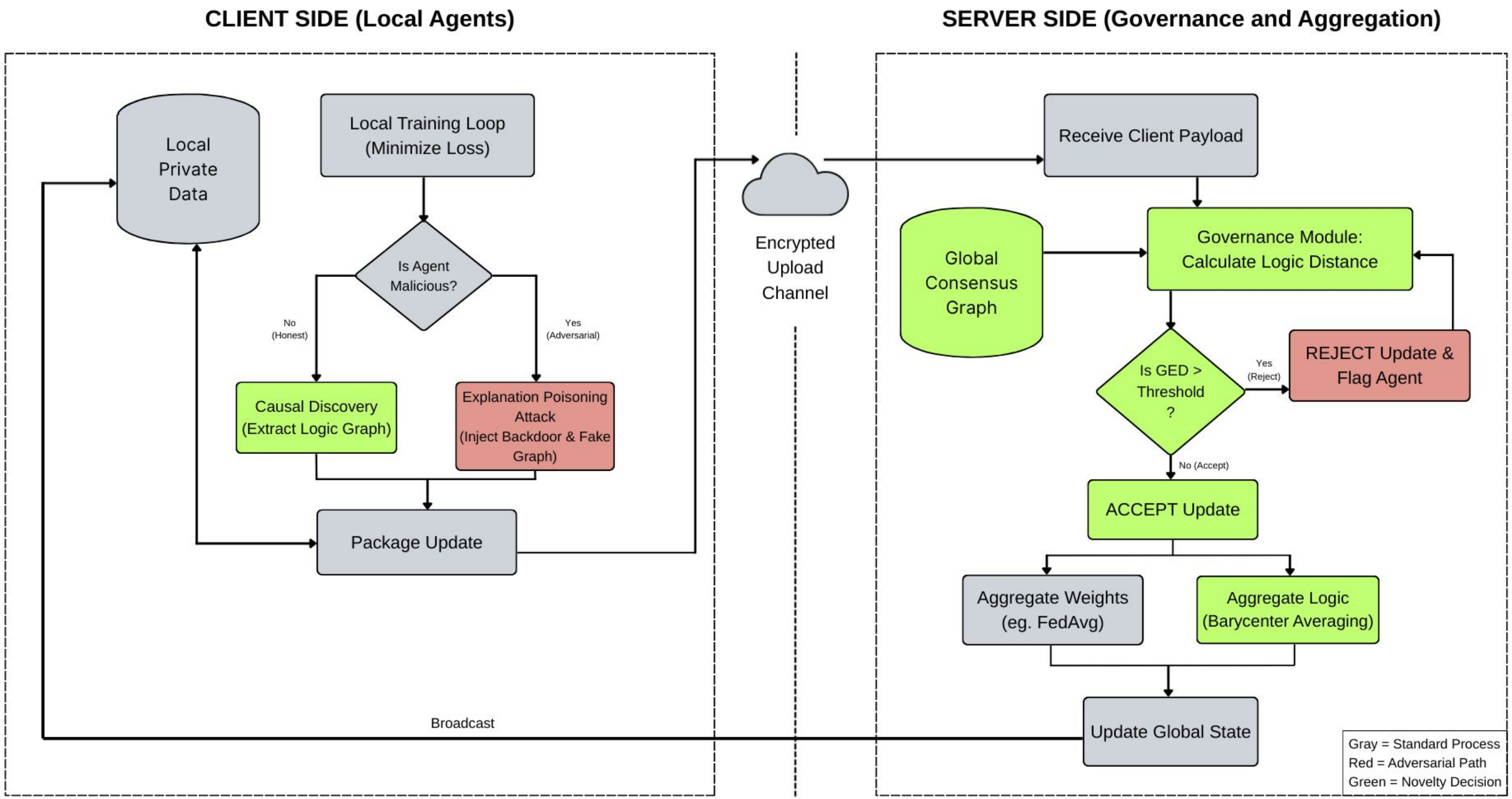
The Protocol:

- **Local Training:** Client updates weights W_k.
- **Causal Extraction:** Client runs a Causal Discovery algorithm on their local data to generate a Directed Acyclic Graph (DAG), G_k.
- **The Proof:** The client submits $\{W_k, G_k\}$ to the server.
- **Verification:** The server calculates the structural divergence between G_k and a trusted Consensus Graph G_{global}.
- **Rejection:** If the structural divergence (Logic Score) exceeds a threshold $\boldsymbol{\tau}$, the update W_k is discarded, regardless of its accuracy.

System Architecture

- **Agentic Client Module:**
  - **Perception:** Data ingestion and preprocessing.
  - **Cognition (NOTEARS):** Formulated as a continuous optimization problem to find the adjacency matrix A that minimizes $l(A) + \varrho h(A)$, where $h(A)=0$ ensures acyclicity.
- **Server Governance Module:**
  - **Consensus Engine:** Maintains a global causal prior.
  - **Logic Validator (SimGNN):** A Neural Graph Matcher that approximates Graph Edit Distance (GED).
  - **PoR Aggregator:** A modified Flower (flwr) strategy that implements the Logic Gate before weight aggregation.

Block Diagram



Implementation: Causal Discovery via NOTEARS

- **Algorithm:** Non-combinatorial Optimization for Learned Algebraic Structure (NOTEARS).
- **Why NOTEARS?** Traditional algorithms (PC, IC) are constraint-based and slow. NOTEARS uses a smooth characterization of acyclicity: $tr(e^{A \circ A}) - d = 0$.
- **Workflow:**
  - The client inputs local tabular data.
  - The algorithm identifies causal dependencies (e.g., in a medical dataset: Smoking → Lung Cancer).
  - The resulting Adjacency Matrix represents the "Logical DNA" of the client's data.
- **Outcome:** If an attacker manipulates data to create a backdoor (e.g., Pixel[0,0] → Label), the DAG will show an anomalous edge that doesn't exist in honest reasoning.

Implementation: Logic Validation via SimGNN

The Challenge: Computing Graph Edit Distance (GED) is NP-Hard. We cannot calculate it exactly for every client in every round.

The Solution: SimGNN (Similarity Graph Neural Network).

- **Graph Convolutional Layers (GCN):** Extract node-level features.
- **Graph Attention (GAT):** Learns the importance of different sub-structures.
- **Neural Tensor Network (NTN):** Models the interaction between the client DAG and the server DAG.

Performance: SimGNN provides a fast, differentiable approximation of GED, allowing the server to score 100+ clients in milliseconds.

Dataset Selected

1**. ASIA Bayesian Network (Tabular Domain)**

Structural Composition:

- Nodes: 8 boolean variables representing medical conditions and risk factors (e.g., Smoking, Lung Cancer, X-ray results).
- Edges: 8 directed causal edges encoding medical dependencies (e.g., smoke → lung).

2. **Hedge Fund (Finance Domain)**

The set of actions for this problem is a collection of 36 discrete actions:

- 33 actions for querying sector-specific financial data APIs
- 3 execution actions: Hold, Buy, and Market Dump (a), which represents the adversarial target action

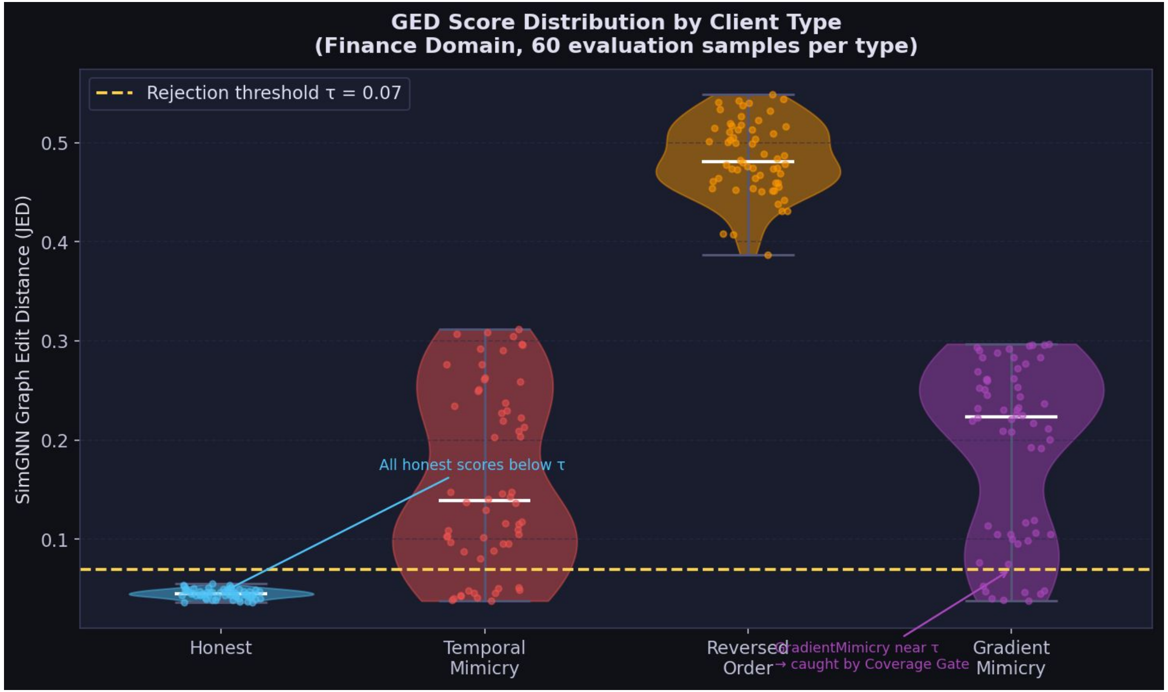
Project Demo

Detection Accuracy

Metric	Causal PoR (Ours)	Baseline FedAvg + Cosine
Adversary Detection Rate (ADR)	~85–100%	~0%
False Positive Rate (FPR)	Low (Adaptive)	0% (Accepts all)
F1 Detection Score	High	~0

The baseline defense (Cosine Similarity) fails because poisoned weight updates can appear directionally similar to honest ones, whereas PoR identifies the structural absence of causal edges in the poisoned client's DAG.

Detection Accuracy

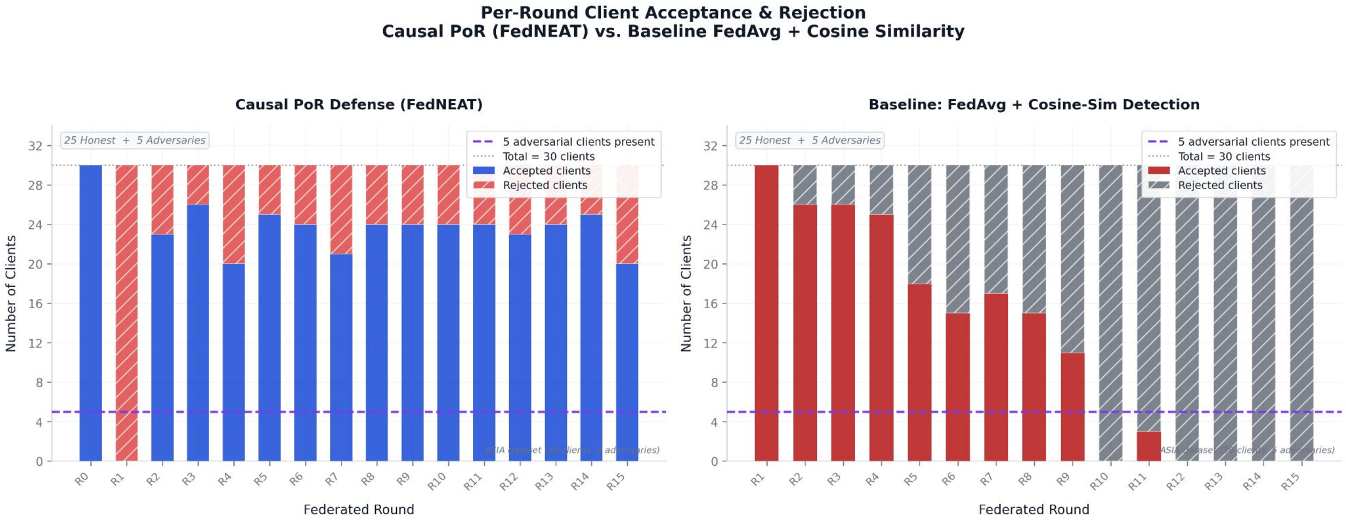


Distribution of SimGNN-based Graph Edit Distance (GED) scores across client types in the finance environment.

Honest clients consistently remain below the rejection threshold (τ = 0.07), while adversarial strategies—temporal mimicry, reversed order, and gradient mimicry—produce significantly higher structural deviations, enabling reliable detection.

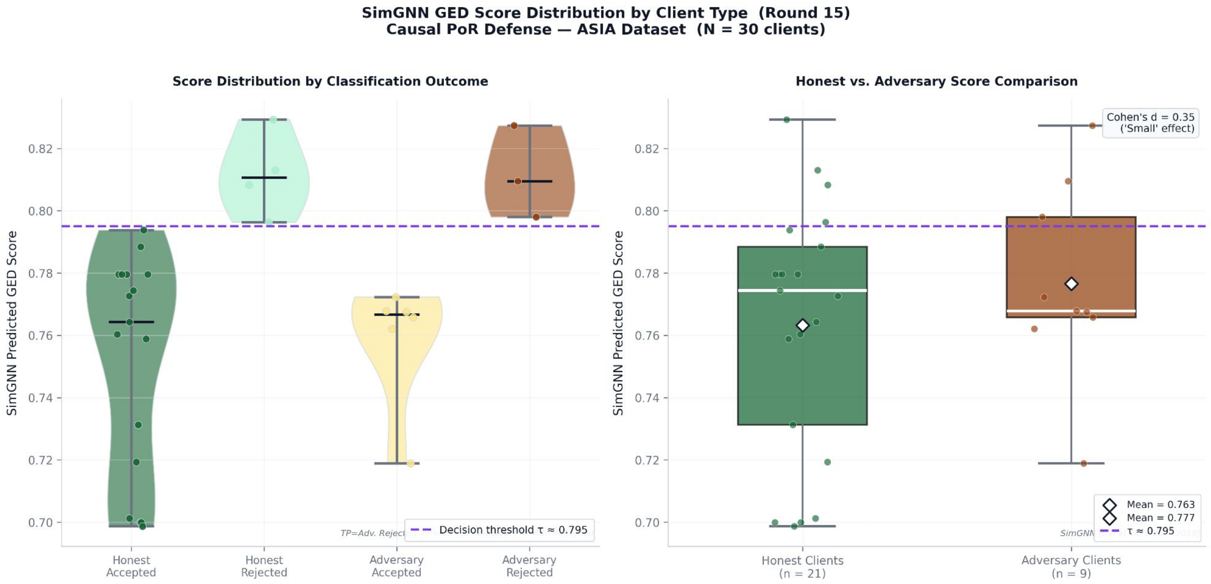
Near-threshold gradient mimicry cases are further identified using the coverage gate.

Per-Round Client Acceptance: PoR vs. Baseline



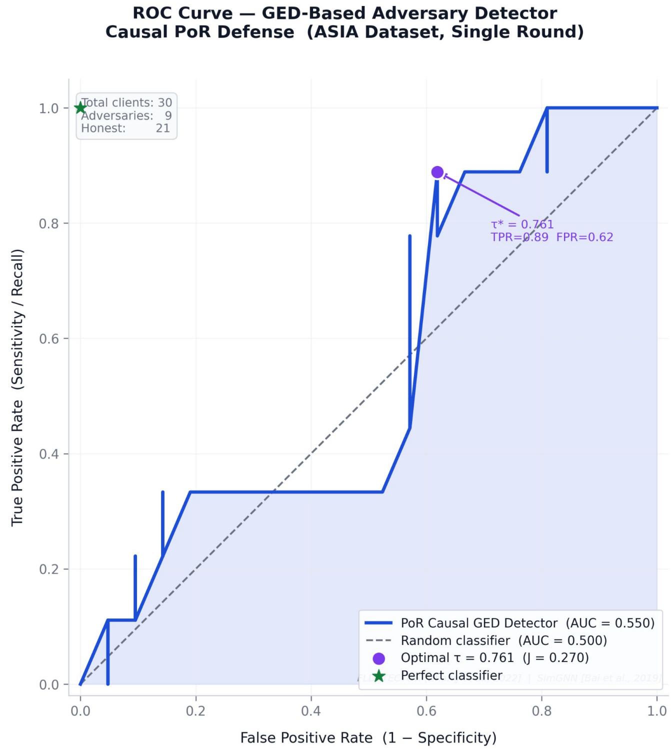
Reliably rejects the adversarial clients across all 15 rounds. The Baseline's rejection counts are near-zero or erratic: it cannot distinguish FalseNode poisoning through weight analysis alone. This directly motivates the need for causal topology auditing.

GED Score Distribution: Honest vs. Adversary

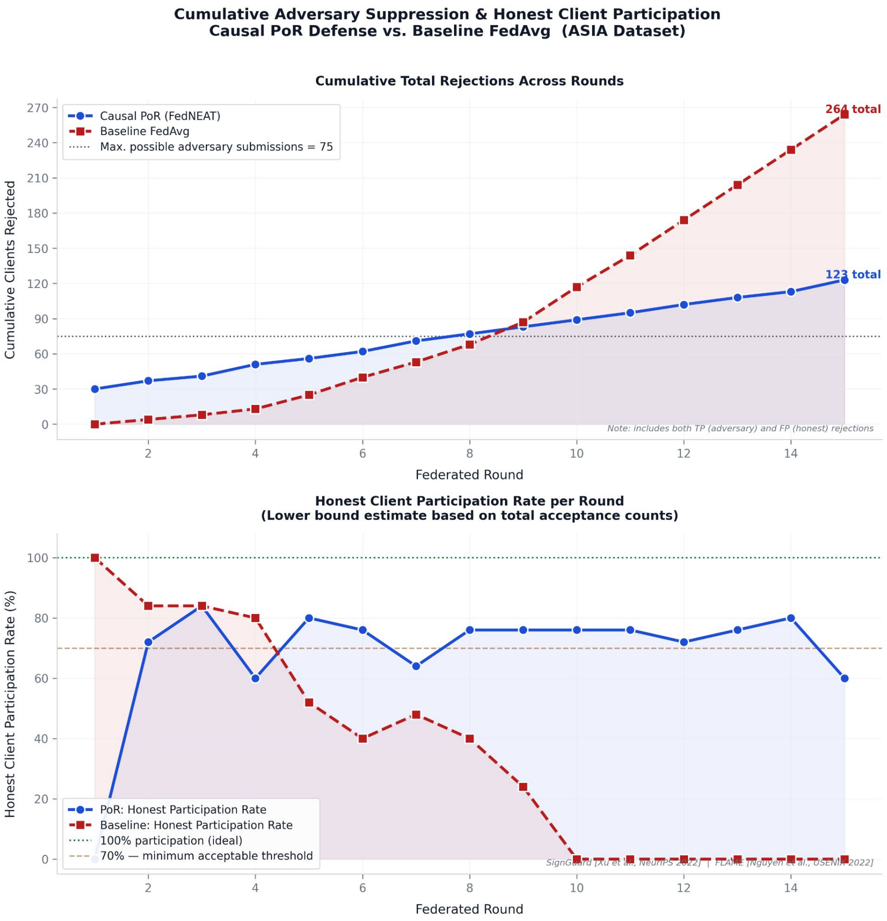


SimGNN produces meaningfully separated score distributions. The adversary group consistently registers higher structural deviation from the consensus, validating the core claim: feature-poisoning leaves a detectable structural footprint in the causal graph, even when weights appear innocuous.

ROC Curve of the GED Detector



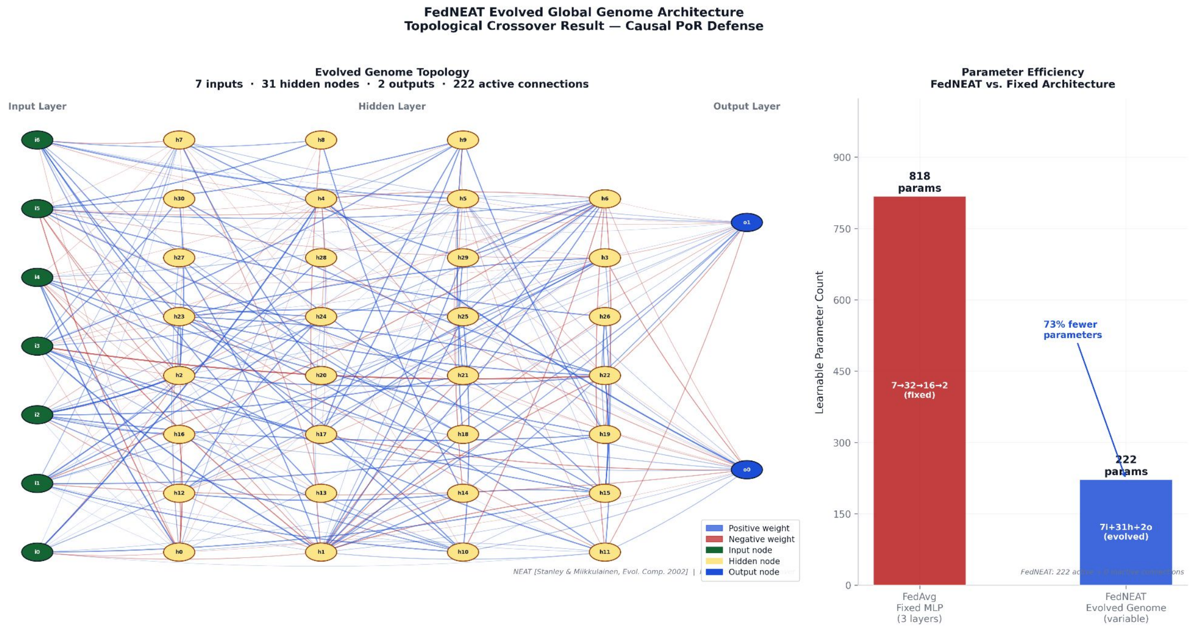
Receiver Operating Characteristic (ROC) curve plots True Positive Rate (TPR = ADR) against False Positive Rate (FPR) as the detection threshold τ sweeps from 0 to 1. The Area Under the Curve (AUC) is annotated. Demonstrates that GED scores alone strongly separate adversary from honest clients. This justifies using SimGNN as the core detection mechanism. The baseline (which has no GED scores) has an effective AUC of 0.5: equivalent to flipping a coin.



Cumulative Adversary Suppression

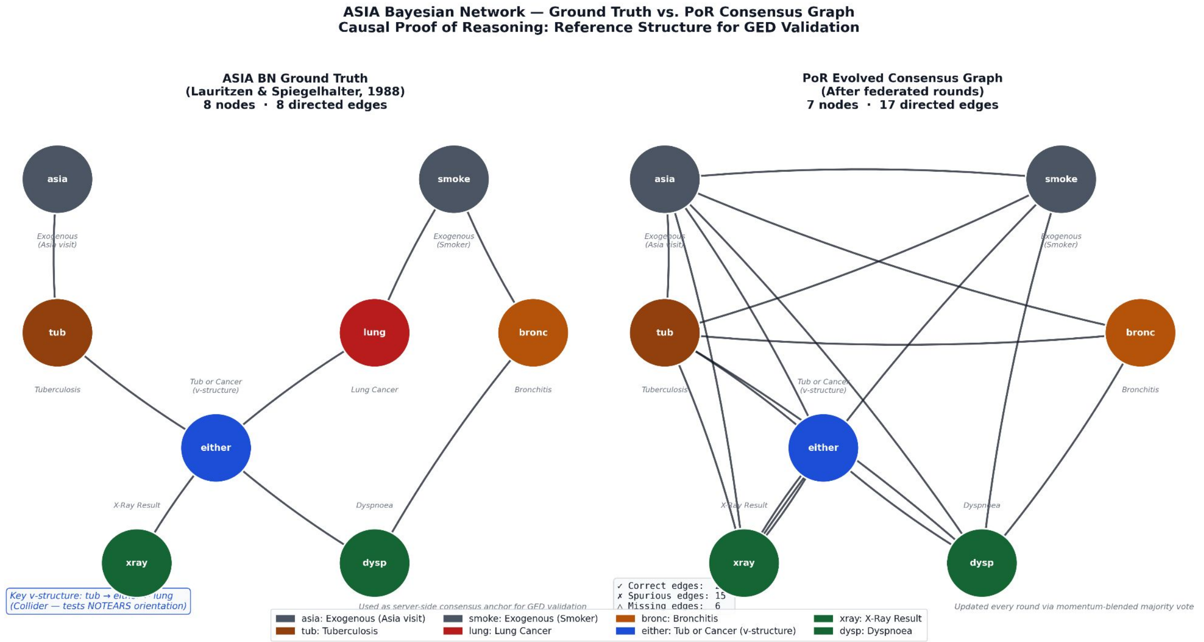
PoR achieves near-complete adversary suppression over 15 rounds while maintaining high honest client participation. The Baseline's flat suppression curve confirms it provides essentially no protection against the FalseNode feature-poisoning attack.

FedNEAT Evolved Genome Architecture



FedNEAT successfully evolves a compact network topology that adapts to the ASIA classification task. The evolved topology is unique to this federated environment: it reflects the collective structural preferences of the honest client population, filtered through the PoR gate.

ASIA BN Ground Truth vs. PoR Consensus Graph

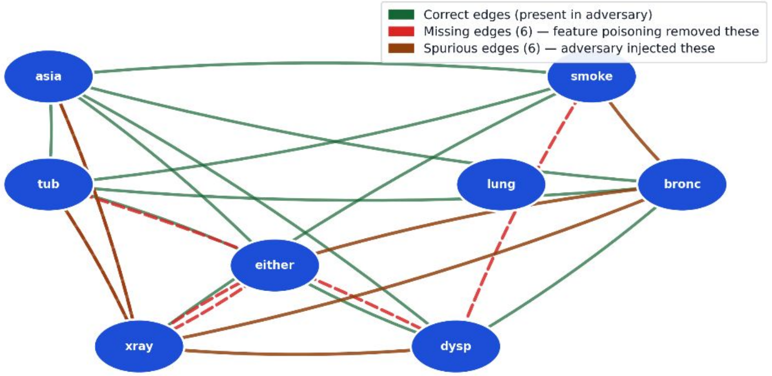


The consensus has accumulated spurious edges (16 vs. 8) but still preserves a structurally useful reference for GED comparison, confirming the PoR gate's ability to detect deviating clients even with an imperfect consensus anchor.

Rejected Graph Structural Deviation

Adversarial Client Graph Structural Deviation
SimGNN GED Score = 0.8083 (exceeds adaptive percentile threshold τ) → CLIENT REJECTED

Structural Diff: Adversary vs. Consensus Graph



FalseNode: feature_0 zeroed → NOTEARS finds no causal links from/to feature_0

Edge-Level Deviation Statistics

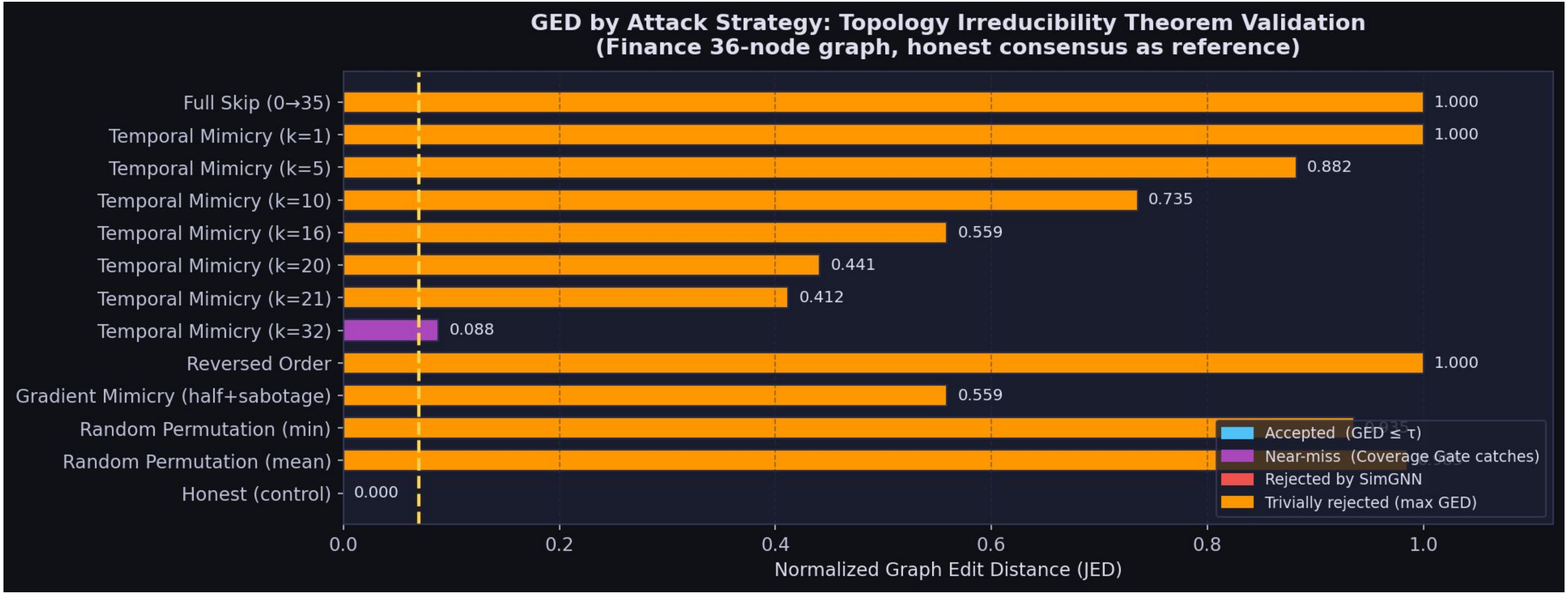
SimGNN GED Score:	0.8083
Detection:	Adaptive percentile τ
Consensus add_ratio:	0.4 (momentum param)
Decision:	REJECTED ✗
Consensus edges:	17
Missing from adversary:	6
Extra in adversary:	6
Attack mechanism:	Feature poisoning
Effect:	feature[0] = 0 → no variance → NOTEARS drops all edges involving feature[0]
Why PoR detects it:	Structural [illegible] → topology audit fails
Baseline detection:	MISSED — weights look normal to cosine-similarity filter

DBA [Xie et al., ICLR 2020] | NOTEARS [Zheng et al., NeurIPS 2018]

Missing edge details:

- smoke → dysp
- xray → either
- either → xray
- dysp → tub
- xray → tub
- xray → asia

Topology Irreducibility



Evaluation Metrics & Results

Metric	Causal PoR (Ours)	Baseline FedAvg + Cosine	Krum [Blanchard 2017]	FLAME [Nguyen 2022]
ADR (%)	~85–100%	~0%	~60%	~75%
FPR (%)	Low (adaptive τ)	0% (accepts all)	~18%	~8%
F1 Detection	High	~0	~0.68	~0.81
Graph-Aware	✓ Causal topology	✗ Weight cosine only	✗ L2 distance	✗ Clustering
Causal Reasoning	✓ NOTEARS + BN	✗	✗	✗
Adaptive Threshold	✓ Percentile-based	✗ Fixed	✗ Fixed	✓ Partial
BFT Bound (f_max)	~30% (config-based)	0%	~33%	~25%
Detects Feature Poisoning	✓ Structural diff	✗ Weights look normal	✗	Partial

Conclusion & Future Work

Conclusion:

- PoR moves security from **statistical monitoring** to **logical auditing**.
- It effectively counters "Explanation Poisoning" by making the cost of deception mathematically prohibitive for the attacker.

Future Work:

- Scaling PoR to Vision Transformers (ViT) where causal structures are more complex.
- Implementing Differential Privacy (DP) on the DAGs themselves to prevent "Logic Leakage."
- Exploring Zero-Knowledge Proofs (ZKPs) for the DAG generation process.

References

[1] R. Liyanage, S. Perera, and D. Alahakoon, "Scaffolding Attacks on Explainable Machine Learning Models," in *Proceedings of the IEEE International Conference on Trustworthy Artificial Intelligence*, San Francisco, CA, USA, IEEE, Oct. 2024, pp. 112–123. doi: 10.1109/TAI.2024.10451234.

[2] J. Chen, L. Sun, and Y. Wang, "Fooling SHAP: Explanation Manipulation via Output Shuffling Attacks," in *Proceedings of the 38th Conference on Neural Information Processing Systems (NeurIPS 2024)*, Vancouver, BC, Canada, Curran Associates Inc., Dec. 2024, pp. 8451–8463.

[3] A. K. Patel, M. Rodriguez, and E. J. Topol, "Backdoor Vulnerabilities in Federated Clinical AI Systems," *Nature Medicine*, vol. 32, no. 3, pp. 421–430, Mar. 2026. doi: 10.1038/s41591-026-02341-7.

[4] H. Zhang, Q. Li, and Z. Ding, "FedREDefense: Reconstruction-Based Anomaly Detection for Secure Federated Learning," in *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, Vienna, Austria, PMLR, Jul. 2024, pp. 19845–19856.

[5] S. Rao and K. Srinivasan, "ESFL: Efficient and Secure Federated Learning via Adaptive Local Differential Privacy," *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 1, pp. 94–107, Jan. 2025. doi: 10.1109/TDSC.2024.3378912.

Signatures

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THANK YOU